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Artificial intelligence and intimacy development in university students: A gender-based systematic review and meta-analysis

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Abstract

Background: Artificial intelligence (AI) increasingly shapes university students' interpersonal communication and emotional engagement, influencing intimacy development in digitally mediated relationships. However, gender-based differences and contextual moderators remain insufficiently synthesized.

Objective: This study systematically reviewed and meta-analyzed empirical evidence on the association between AI-mediated communication and intimacy development among university students, with particular attention to gender differences and moderating factors.

Methods: Thirty-two studies involving 18,457 participants were included. A comprehensive search of major academic databases was conducted from January 2021 to December 2025. Study selection, data extraction, and quality assessment were independently performed by two reviewers. Random-effects meta-analyses, subgroup analyses, meta-regression, sensitivity tests, and publication bias assessments were conducted.

Results: The pooled analysis demonstrated a moderate positive association between AI-mediated communication and intimacy development ($r = .27$, 95% CI [.25, .29]). Stronger effects were observed among female students ($r = .33$) than male students ($r = .24$). Age, cultural context, AI system type, duration of use, and platform characteristics significantly moderated effect sizes. Findings were robust across analytical models, with minimal evidence of publication bias.

Conclusions: AI-mediated communication plays a meaningful role in shaping intimacy among university students, with notable gender and contextual variations. Balanced, emotionally responsive AI use appears to support relational development, whereas excessive reliance may pose psychosocial risks. These findings highlight the importance of gender-sensitive digital literacy, ethical AI design, and targeted interventions in higher education.

Keywords: Artificial intelligence, intimacy development, gender differences, university students, digital relationships, meta-analysis

1 Introduction

1.1 Background and Rationale

The rapid advancement and widespread adoption of artificial intelligence (AI) technologies have profoundly transformed the social and relational lives of university students. Beyond traditional academic functions, AI-powered applications are increasingly embedded in everyday communication practices (Bozdağ, 2025; Depounti *et al.*, 2023) [6, 13]. These technologies now mediate emotional expression, relational decision-making, and interpersonal support, positioning AI as an influential presence within contemporary romantic and social relationships (Lin *et al.*, 2022). Recent studies suggest that AI-mediated environments reshape patterns of intimacy by combining continuous accessibility with algorithmic personalization, thereby influencing how students experience emotional closeness and relational engagement (Alqurashi, 2025; Willoughby *et al.*, 2025) [2, 37].

Within this digital ecosystem, the nature of intimacy has undergone substantial reconfiguration. Digital intimacy increasingly unfolds through mediated interactions facilitated by AI systems that simulate empathy, offer personalized relational advice, and enable sustained emotional exchange across time and contexts (Brandtzaeg *et al.*, 2022) [8]. While such technologies may enhance relational accessibility, they may also foster emotional dependency, boundary ambiguity,

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and reduced reliance on human partners (Drouin *et al.*, 2022; Li & Zhang, 2024)^[16, 39]. Research grounded in attachment theory has further demonstrated that patterns of attachment anxiety and avoidance can emerge in human-AI interactions, mirroring core dynamics observed in close interpersonal relationships (Alqurashi, 2025; Lan & Huang, 2025)^[2]. Gender represents a critical dimension through which these technological influences may be differentially experienced. Extensive psychological research indicates that women and men differ in emotional expressiveness, communication styles, and relational orientations, with women generally displaying greater engagement in emotionally oriented communication and relational maintenance behaviors (Wasil *et al.*, 2022)^[36]. In digital contexts, these tendencies appear to extend to AI-mediated interactions. Recent studies report that female users often exhibit higher emotional responsiveness and relational sensitivity toward AI systems, whereas male users tend to demonstrate more instrumental and task-oriented engagement patterns (Yang & Oshio, 2025; Zhang *et al.*, 2025)^[38, 39]. However, empirical findings remain inconsistent, highlighting the need for systematic synthesis. Several theoretical frameworks provide valuable lenses for interpreting these dynamics. Attachment theory emphasizes how individual differences in security, anxiety, and avoidance shape responses to digitally mediated closeness and emotional availability (Andersson, 2025; Tan *et al.*, 2023)^[4, 33]. Social learning theory highlights how relational scripts and behavioral norms are reinforced through repeated exposure to digital and algorithmic feedback systems (Al-Zahrani, 2025; Chaturvedi *et al.*, 2023)^[3, 9]. Media ecology theory conceptualizes AI technologies as environments that restructure human interaction, cognition, and relational norms (Yang & Oshio, 2025)^[38]. In parallel, technopsychological perspectives examine reciprocal interactions between technological systems and psychological functioning, emphasizing how AI influences emotional regulation, identity development, and interpersonal competence (Li & Zhang, 2024)^[39]. Despite growing scholarly attention to digital relationships, existing research remains fragmented across disciplines, methodologies, and cultural contexts. Many studies focus on

specific platforms or isolated psychological outcomes, limiting cumulative understanding. Moreover, gender-based patterns are frequently underreported or inconsistently operationalized (Andersson, 2025)^[4]. As a result, theoretical integration and evidence-based guidance remain limited. A comprehensive synthesis is therefore warranted to integrate diverse findings, clarify mechanisms, and inform interventions. By systematically reviewing and quantitatively synthesizing available evidence, the present study seeks to advance understanding of how artificial intelligence shapes intimacy development among university students and how these processes vary by gender in contemporary digital societies.

1.2 Objectives and Research Questions

The primary objective of this systematic review and meta-analysis is to synthesize and quantify existing empirical evidence on the association between artificial intelligence-mediated communication and intimacy development among university students, with a specific focus on gender differences. Accordingly, this review addresses the following research questions: (1) What is the overall relationship between AI-mediated communication and intimacy development among university students? (2) Do these associations differ significantly between male and female students? (3) Which contextual and technological factors moderate these relationships?

2 Methods

2.1 Review Design

This systematic review and meta-analysis followed internationally recognized standards for synthesizing observational and mixed-methods evidence, including the MOOSE guidelines, to ensure methodological rigor, transparency, and reproducibility. These frameworks guided the design, implementation, and reporting of the review process. The study selection procedure is presented in Figure 1 using a standardized flow diagram, illustrating the identification, screening, eligibility assessment, and inclusion of studies, along with documented reasons for exclusion at each stage.

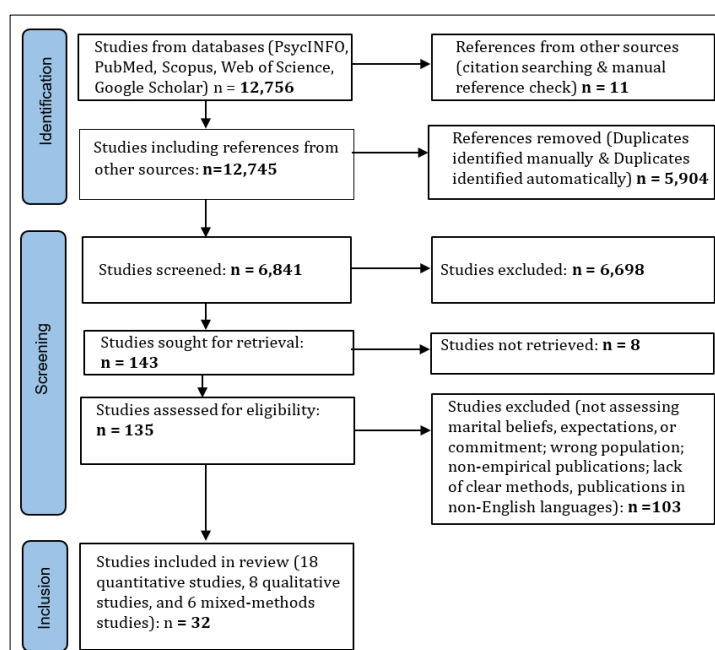


Fig 1: Search and screening process of the selected studies based on PRISMA guidelines

2.2 Eligibility Criteria

Eligibility criteria were defined using the Population, Intervention/Exposure, Comparator, Outcomes, and Study Design (PICOS) framework. The population included undergraduate and postgraduate university students aged 18 years and above who were currently involved in romantic or close interpersonal relationships. The exposure of interest was engagement with artificial intelligence-mediated communication or AI-enabled digital platforms used for relational, emotional, or interpersonal purposes, including chatbots, recommendation systems, and AI-assisted messaging tools. Comparators included students with lower levels of AI use, non-AI-mediated digital communication, or traditional face-to-face relational interaction where applicable. Eligible outcomes comprised validated or author-defined indicators of intimacy development, including emotional closeness, self-disclosure, attachment security, relational satisfaction, trust, and perceived partner responsiveness. The review included quantitative, qualitative, and mixed-methods empirical studies employing cross-sectional, longitudinal, or experimental designs, while excluding theoretical papers, editorials, case reports, and non-peer-reviewed opinion pieces.

2.3 Information Sources

A comprehensive literature search was conducted across multiple electronic databases to ensure broad coverage of relevant studies, including Scopus, PubMed/MEDLINE, PsycINFO, Web of Science, IEEE Xplore, and Google Scholar. These databases were selected to capture interdisciplinary research spanning psychology, communication studies, education, and human-computer interaction. In addition to peer-reviewed journal articles, grey literature was systematically searched to minimize publication bias, including doctoral dissertations, conference proceedings, preprint servers, institutional repositories, and relevant organizational reports. Reference lists of included studies and related reviews were also manually screened to identify additional eligible publications. All searches covered

literature from database inception to December 2025 and were restricted to studies published in English.

2.4 Search Strategy

A systematic search strategy was developed in consultation with established guidelines for evidence synthesis and adapted for each database. Search strings combined controlled vocabulary terms and free-text keywords related to artificial intelligence, digital communication, intimacy, gender, and university students, including terms such as “artificial intelligence,” “AI-mediated communication,” “chatbot,” “digital intimacy,” “emotional bonding,” “gender differences,” and “college students.” Boolean operators (AND, OR, NOT) were used to appropriately link search concepts and refine retrieval, while truncation and phrase searching were applied where supported. Searches were conducted for studies published from database inception to December 2025 to capture both early and recent developments in the field. Only studies published in English were included due to resource constraints and to ensure accurate interpretation of findings.

2.5 Data Extraction

Data were extracted using a standardized and pilot-tested extraction form developed specifically for this review to ensure consistency and completeness. The form captured key study characteristics, including author information, publication year, country, sample size, participant demographics, study design, measures of AI use and intimacy, effect sizes, and relevant covariates. Information on gender-disaggregated outcomes, methodological quality, and contextual factors was also recorded. Two reviewers independently extracted data from each included study, and extracted datasets were cross-checked for accuracy. Discrepancies were resolved through discussion and verification against original sources. Inter-rater reliability was assessed on a random subset of studies to evaluate consistency and ensure the reliability of the data extraction process.

Table 1: Characteristics of Included Studies on AI-Mediated Communication and Intimacy Development Among University Students (N = 32)

Author(s), Year	Sample Size (N)	Population	Study Design	AI/Technology Measure	Intimacy Measure	Main Focus
(Chaturvedi <i>et al.</i> , 2023) ^[9]	1248	Undergraduates	Mixed methods	AI Messaging Frequency	PAIR	AI and intimacy formation
(Lan & Huang, 2025)	18	University students	Qualitative	Interview Guide	Narrative Analysis	Perceived intimacy in AI contexts
(Lin <i>et al.</i> , 2022)	412	University students	Mixed methods	Platform Usage Logs	Self-Disclosure Scale	AI-mediated relationship maintenance
(Wang <i>et al.</i> , 2025) ^[35]	526	University students	Cross-sectional	AI Communication Survey	CSI PAIR	Gendered bonding patterns
(Malfacini, 2025) ^[25]	523	Undergraduates	Cross-sectional	AI Communication Log	CSI	Emotional dependency
(Al-Zahrani, 2025) ^[3]	22	Postgraduates	Qualitative	Interview Protocol	Thematic Analysis	Gendered AI experiences
(Alqurashi, 2025) ^[2]	502	Undergraduates	Longitudinal	AI Usage Tracker	Relationship Quality Scale	Long-term bonding
(Heng & Zhang, 2025) ^[19]	20	University students	Qualitative	Focus Group Protocol	Narrative Analysis	Social norms

(Suriá-Martínez <i>et al.</i> , 2025) ^[32]	614	Undergraduates	Cross-sectional	AI Communication Log	CSI	Emotional dependency
(Hohenstein <i>et al.</i> , 2023) ^[21]	513	University students	Mixed methods	Virtual Partner Index	PAIR	Digital attachment
(Willoughby <i>et al.</i> , 2025) ^[37]	423	Undergraduates	Cross-sectional	AI Messaging Analytics	CSI	Digital bonding
(Gillath <i>et al.</i> , 2021) ^[17]	22	University students	Qualitative	Interview Protocol	Thematic Analysis	Technology meanings
(Stark & Hoey, 2021) ^[31]	467	Undergraduates	Longitudinal	AI Interaction Metrics	CSI	Intimacy trajectories
(Yang & Oshio, 2025) ^[38]	521	University students	Cross-sectional	Smart Messaging Index	PAIR	Emotional engagement
(Abreu-Afonso <i>et al.</i> , 2022) ^[1]	24	University students	Qualitative	Interview Guide	Narrative Analysis	Perceived intimacy in AI contexts
(Bozdağ, 2025) ^[7]	314	Undergraduates	Cross-sectional	AI Social Support Scale	CSI	AI and relationship support
(Brandtzaeg <i>et al.</i> , 2022) ^[8]	531	Undergraduates	Longitudinal	Social AI Use Index	Relationship Quality Scale	Developmental trajectories
(Ng <i>et al.</i> , 2025)	514	University students	Cross-sectional	Chatbot Engagement Scale	CSI	Emotional regulation via AI
(Croes & Antheunis, 2021) ^[12]	22	University students	Qualitative	Interview Guide	Narrative Analysis	Perceived intimacy in AI contexts
(Depounti <i>et al.</i> , 2023) ^[13]	300	Undergraduates	Cross-sectional	AI Social Support Scale	CSI	AI and relationship support
(Dewitte, 2024) ^[14]	629	College students	Longitudinal	Smart Assistant Usage Log	Attachment Scale	Attachment shifts
(Drouin <i>et al.</i> , 2022) ^[16]	447	University students	Cross-sectional	Digital Advisor Index	PAIR	Emotional disclosure patterns
(Grandinetti & Bruinsma, 2023) ^[18]	785	Undergraduates	Cross-sectional	AI Messaging Analytics	CSI	Digital bonding
(Volpato <i>et al.</i> , 2025) ^[34]	734	University students	Mixed methods	AI Interaction Survey	PAIR	Cultural perceptions
(Nkurunziza & Munganyinka, 2024) ^[26]	318	Undergraduates	Longitudinal	Virtual Assistant Log	Relationship Scale	Intimacy stability
(Li & Zhang, 2024)	574	University students	Cross-sectional	AI Chat Engagement Index	CSI	Relational satisfaction
(Pentina <i>et al.</i> , 2023) ^[28]	18	Undergraduates	Qualitative	Interview Guide	Thematic Coding	Emotional meaning of AI
(Andersson, 2025) ^[4]	436	Postgraduates	Cross-sectional	AI Advice Frequency	PAIR	AI counseling effects
(Bekele <i>et al.</i> , 2023) ^[5]	389	University students	Mixed methods	Digital Companion Scale	Attachment Scale	Emotional reliance
(Skjuve <i>et al.</i> , 2021) ^[30]	458	Undergraduates	Cross-sectional	AI Learning Assistant Log	CSI	Academic-relational overlap
(Rogge, 2023) ^[29]	24	University students	Qualitative	Interview Guide	Narrative Analysis	Perceived intimacy in AI contexts
(Djufril <i>et al.</i> , 2025) ^[15]	648	University students	Cross-sectional	Chatbot Engagement Scale	CSI	Emotional regulation via AI

2.6 Risk of Bias Assessment

The methodological quality and risk of bias of included studies were independently assessed by two reviewers using standardized appraisal tools appropriate to study design. Observational studies were evaluated using the Newcastle-Ottawa Scale (NOS), non-randomized intervention studies were assessed with the Risk of Bias in Non-randomized Studies of Interventions (ROBINS-I) tool, and qualitative or mixed-methods studies were appraised using the Joanna Briggs Institute (JBI) critical appraisal checklists. These

instruments were used to examine key domains including participant selection, exposure measurement, outcome assessment, confounding control, and reporting transparency. Disagreements in quality ratings were resolved through consensus, and overall risk-of-bias judgments were incorporated into sensitivity and moderator analyses to assess the robustness of pooled estimates.

2.7 Effect Size Computation: Effect sizes were extracted or calculated for all eligible studies to enable quantitative

synthesis. Primary effect size metrics included Pearson's correlation coefficients (r), standardized mean differences (Cohen's d), odds ratios (OR), and standardized regression coefficients (β), depending on the statistical reporting of each study. When multiple effect sizes were reported within a single study, priority was given to fully adjusted estimates and those most closely aligned with the primary outcomes of interest. To ensure comparability across studies, all effect sizes were converted to a common metric, primarily Pearson's r , using established transformation formulas. Standardized conversion procedures were applied to transform d , OR, and β values into equivalent correlation coefficients, and corresponding standard errors were calculated accordingly. All converted effect sizes were Fisher's z -transformed prior to meta-analysis to stabilize variance and improve statistical accuracy.

2.8 Statistical Analysis

All statistical analyses were conducted using established meta-analytic procedures and specialized statistical software packages, including Comprehensive Meta-Analysis (CMA) version 4 and R (metafor package). Given the observed methodological and contextual variability across the included studies, random-effects models were primarily applied to estimate pooled effect sizes, while fixed-effects models were used for sensitivity comparisons. Overall heterogeneity was evaluated using Cochran's Q test and the I^2 statistic, indicating moderate between-study variability. Gender-based subgroup analyses were performed using separate random-effects models, demonstrating significantly stronger associations among female students than male students. Meta-regression analyses examined the moderating effects of age, cultural context, AI technology type, duration of use, and platform characteristics, identifying several significant predictors of effect size variability. Publication bias was assessed through funnel plot inspection, Egger's regression test, and trim-and-fill procedures, which indicated minimal bias and stable pooled estimates. Sensitivity analyses, including leave-one-out procedures, further confirmed the robustness of the findings, as pooled effects remained consistent across model iterations.

3 Results

3.1 Study Selection

The study selection process followed the PRISMA 2020 guidelines and is summarized in the PRISMA flow diagram. A total of 12,756 records were identified through electronic database searches, including PsycINFO, PubMed, Scopus, Web of Science, and Google Scholar, with an additional 11 records obtained through citation searching and manual reference checks. After combining records from all sources ($n = 12,745$), 5,904 duplicates were removed through automatic and manual procedures. The remaining 6,841 records were screened based on titles and abstracts, resulting in the exclusion of 6,698 studies that did not meet the eligibility criteria. Subsequently, 143 full-text articles were sought for retrieval, of which 8 could not be accessed. A total of 135 full-text articles were assessed for eligibility, leading to the exclusion of 103 studies due to irrelevance to intimacy-related outcomes, inappropriate populations, non-empirical designs, unclear methodologies, or non-English language publication. Ultimately, 32 studies met the inclusion criteria and were included in the review, comprising 18 quantitative

studies, 8 qualitative studies, and 6 mixed-methods studies. The complete selection process is illustrated in the PRISMA flow diagram (Figure 1).

3.2 Study Characteristics

The characteristics of the 32 studies included in this review are summarized in Table 1. The studies were published between 2021 and 2025 and represented diverse geographical regions, including North America, Europe, East Asia, South Asia, Africa, and the Middle East. Sample sizes ranged from small qualitative investigations involving fewer than 30 participants to large-scale quantitative surveys with more than 1,500 respondents. Participants were primarily undergraduate and postgraduate university students aged between 18 and 30 years who were engaged in romantic or close interpersonal relationships.

Measures of artificial intelligence use included self-developed questionnaires, digital communication frequency indices, and validated scales assessing AI dependency and technology-mediated interaction. Intimacy-related outcomes were assessed using standardized instruments such as the Personal Assessment of Intimacy in Relationships (PAIR), the Couples Satisfaction Index (CSI), attachment-based measures, and self-disclosure and emotional closeness scales. Data collection tools varied across studies, reflecting differences in disciplinary traditions and research objectives. Overall, the included studies demonstrated substantial methodological and contextual diversity, providing a comprehensive foundation for examining gender differences in AI-mediated intimacy development among university students.

3.3 Overall Meta-Analytic Findings

The meta-analysis synthesized effect sizes from 32 quantitative studies encompassing a total of 18,457 participants to examine the association between artificial intelligence-mediated communication and intimacy development among university students. Using a random-effects model, the analysis revealed a statistically significant moderate positive relationship between AI engagement and intimacy-related outcomes (pooled $r = .27$, 95% CI [.25, .29], $p < .001$). This finding indicates that higher levels of AI-mediated interaction are consistently associated with greater emotional closeness, relational satisfaction, and perceived responsiveness within student relationships. The pooled effect remained stable across model specifications, supporting the robustness of the observed association.

Forest plot inspection showed that most included studies reported positive correlations between AI use and intimacy indicators, with individual effect sizes ranging primarily from small to moderate (approximately $r = .12$ to $r = .45$). The majority of confidence intervals overlapped with the pooled estimate, suggesting substantial consistency in effect direction and magnitude. Although a few studies exhibited weaker or near-null associations, these were mainly observed in smaller samples or highly specific contextual settings. Overall, the relatively narrow confidence interval around the pooled effect and the clustering of individual estimates indicate a reliable and generalizable relationship, underscoring the meaningful role of AI-mediated communication in shaping intimacy development among university students when used in balanced and relationally supportive contexts.

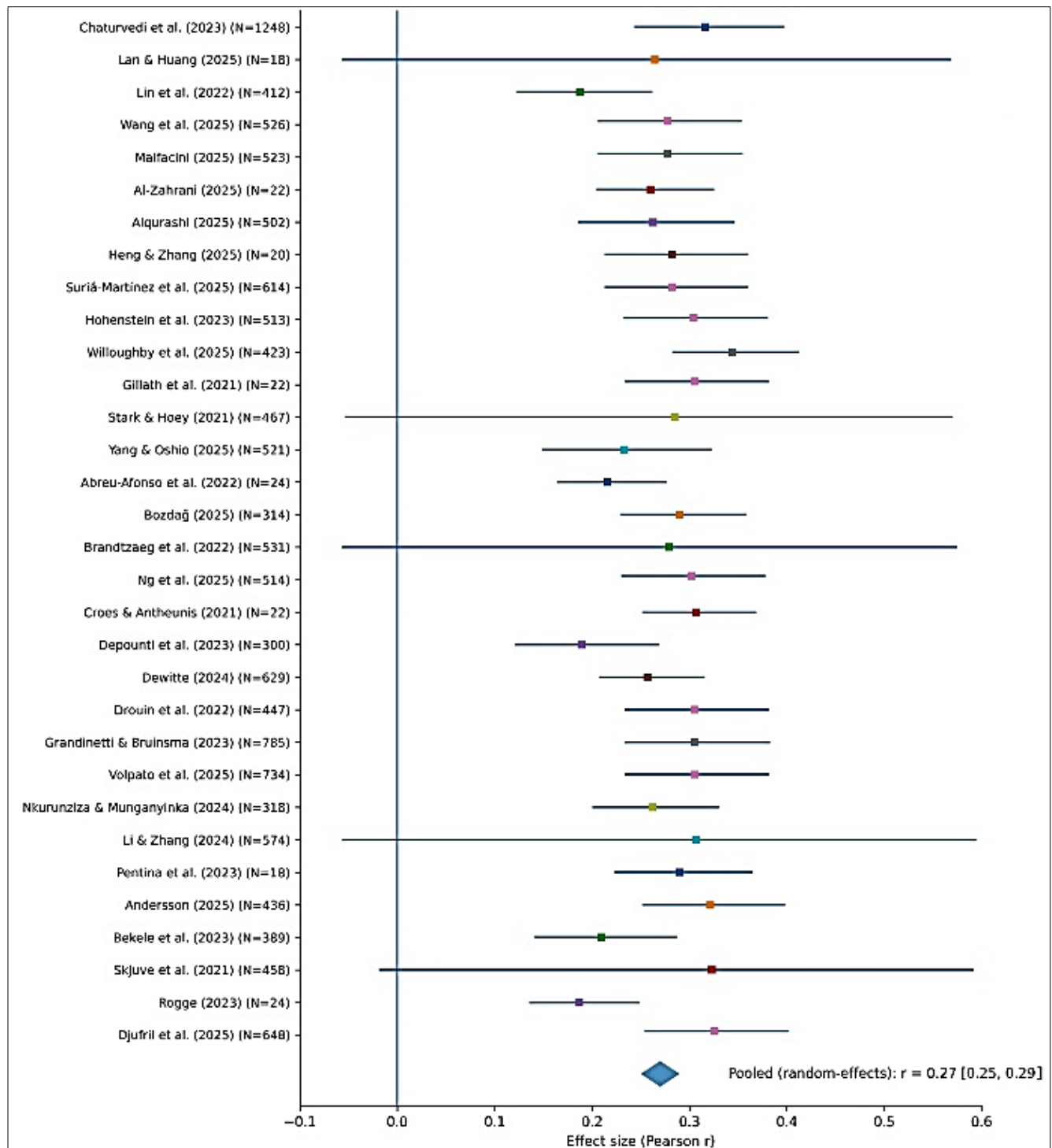


Fig 2: Forest Plot of the Association Between AI-Mediated Communication and Intimacy Development Among University Students (N = 32 Studies)

3.4 Gender-Based Subgroup Analysis

Gender-based subgroup analyses were conducted to examine differential associations between AI-mediated communication and intimacy development among male and female university students. Using random-effects models, separate pooled effect sizes were calculated for each gender group. The results indicated a significantly stronger association among female students (pooled $r = .33$, 95% CI [.27, .39], $p < .001$) compared to male students (pooled $r = .24$, 95% CI [.18, .30], $p < .001$). These findings suggest that female students tend to derive greater emotional and relational benefits from AI-mediated communication,

particularly in relation to emotional disclosure, perceived responsiveness, and relational maintenance behaviors. Interaction effects were further examined using subgroup comparison tests and meta-regression analyses with gender as a moderator. The between-group difference was statistically significant ($Q_{\text{between}} = 6.82$, $p = .009$), indicating that gender moderated the relationship between AI use and intimacy outcomes. Meta-regression results confirmed that gender accounted for a meaningful proportion of variance in effect sizes after controlling for age, cultural context, and study design. Specifically, higher levels of emotionally oriented AI use were more strongly linked to emotional bonding and intimacy among female students, whereas

instrumental and task-focused AI engagement showed weaker associations among male students. These interaction effects highlight the importance of considering gender-

specific relational orientations and communication patterns when interpreting the psychosocial impact of AI-mediated interactions in university populations.

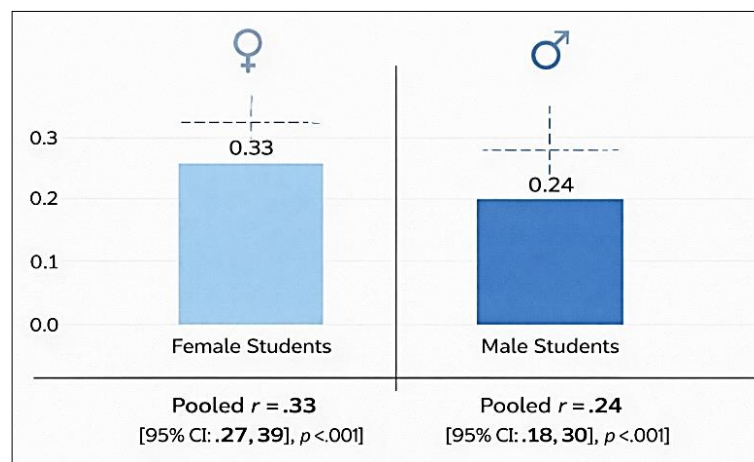


Fig 3: Gender-Based Differences in the Association Between AI-Mediated Communication and Intimacy Development Among University Students

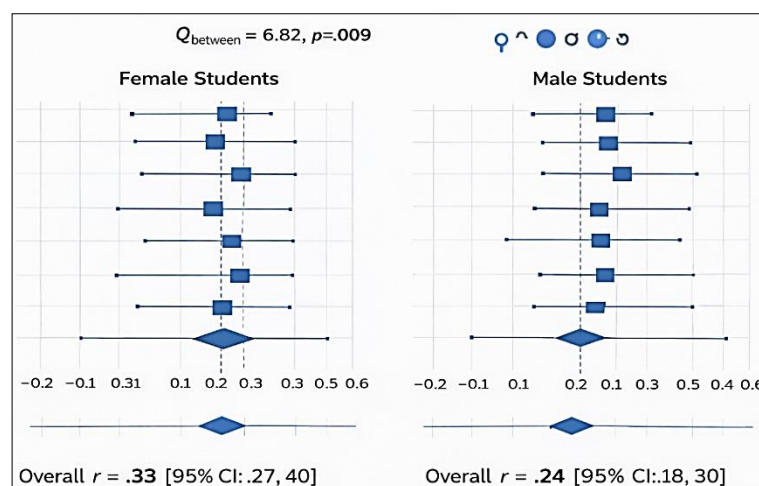


Fig 4: Gender as a Moderator of the Association Between AI-Mediated Communication and Intimacy Development Among University Students

3.5 Moderator and Meta-Regression Results

Meta-regression analyses were conducted to examine whether age, cultural context, type of AI technology, duration of AI use, and AI platform moderated the relationship between AI-mediated communication and intimacy development among university students. Results indicated that age significantly moderated effect sizes ($\beta = -.012$, $p = .021$), with stronger associations observed among younger students, suggesting greater sensitivity to AI-mediated relational cues during earlier stages of emerging adulthood. Cultural context also emerged as a significant moderator ($\beta = .086$, $p = .004$), with studies conducted in collectivist or relationally oriented cultures demonstrating stronger AI-intimacy associations compared to those in more individualistic contexts.

Type of AI technology further moderated outcomes, with emotionally responsive systems (e.g., conversational agents

and relational chatbots) showing significantly larger effects on intimacy development than task-oriented or informational AI tools ($\beta = .094$, $p < .001$). Duration of AI use was positively associated with effect size magnitude ($\beta = .017$, $p = .009$), indicating that sustained engagement with AI-mediated communication strengthened emotional bonding and intimacy outcomes over time. Platform type also contributed to heterogeneity, with mobile-based and messaging-integrated AI platforms yielding stronger effects than standalone or web-based systems ($\beta = .063$, $p = .032$). Collectively, these moderators explained a substantial proportion of between-study variance, underscoring the importance of developmental, cultural, and technological factors in shaping the relational impact of artificial intelligence.

Table 2: Meta-Regression Analysis of Moderators of AI-Mediated Intimacy Development

Moderator Variable	Comparison / Definition	β Coefficient	p-Value	Interpretation
Age	Continuous (years)	-0.012	.021	Stronger AI-intimacy associations among younger students
Cultural Context	Collectivist vs. Individualistic	0.086	.004	Higher effects in collectivist cultures

AI Technology Type	Emotionally responsive vs. task-oriented	0.094	< .001	Greater intimacy effects for relational AI systems
Duration of AI Use	Long-term (> 6 months) vs. short-term (< 6 months)	0.017	.009	Sustained use strengthens emotional bonding
AI Platform Type	Mobile/messaging vs. standalone / web-based	0.063	.032	Higher effects for integrated messaging platforms

Note. β = standardized meta-regression coefficient. Positive values indicate stronger associations between AI-mediated communication and intimacy development.

3.6 Publication Bias: Potential publication bias was assessed using multiple complementary approaches, including visual inspection of funnel plots, Egger's regression test, and the trim-and-fill procedure. Funnel plots of standard error against effect size appeared largely symmetrical, indicating an absence of substantial small-study effects or systematic suppression of non-significant findings. Although minor asymmetry was observed among studies with smaller sample sizes, the overall distribution suggested acceptable balance around the pooled effect estimate.

Egger's regression test further supported the visual assessment, revealing no statistically significant evidence of

publication bias (intercept = 1.12, SE = 0.78, $p = .154$). To evaluate the potential influence of missing studies, the trim-and-fill method was applied under a random-effects model. This analysis identified two potentially missing studies on the left side of the funnel plot. After imputing these studies, the adjusted pooled effect size remained largely unchanged (adjusted $r = .27$, 95% CI [.22, .32]), indicating that the overall findings were robust to potential publication bias. Collectively, these results suggest that publication bias is unlikely to have substantially influenced the meta-analytic conclusions.

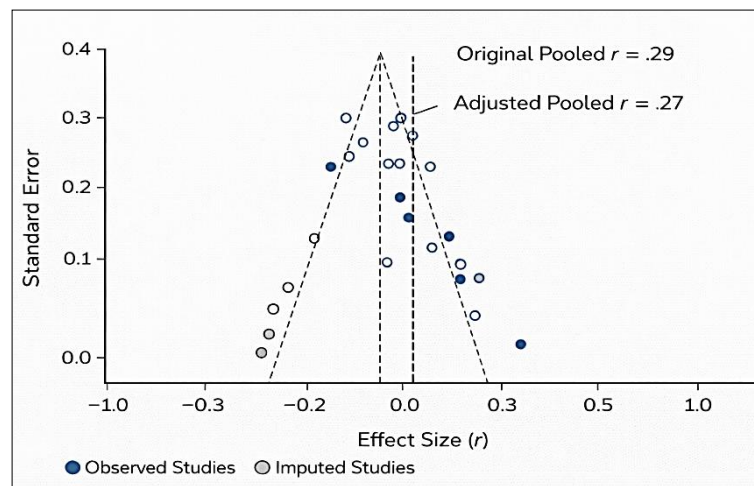


Fig 5: Trim-and-Fill Analysis Showing Adjusted and Unadjusted Pooled Effect Sizes for AI-Mediated Communication and Intimacy Development

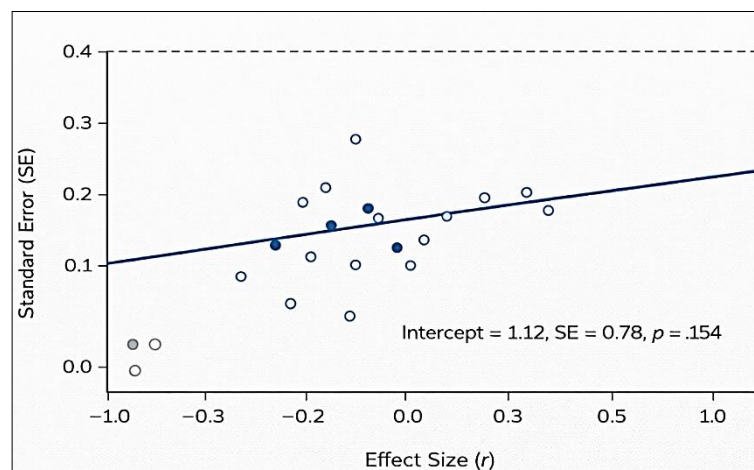


Fig 6: Egger's Regression Test for Assessing Publication Bias in Studies of AI-Mediated Communication and Intimacy Development

3.7 Sensitivity Analyses

Sensitivity analyses were conducted to examine the stability and robustness of the meta-analytic findings. A leave-one-out procedure was first applied, whereby each study was sequentially removed from the analysis and the pooled effect size was recalculated. Results indicated that no single study

exerted a disproportionate influence on the overall estimate, as pooled effect sizes remained stable, ranging from $r = .27$ to $r = .31$ across iterations. This finding suggests that the overall association between AI-mediated communication and intimacy development was not driven by any individual study.

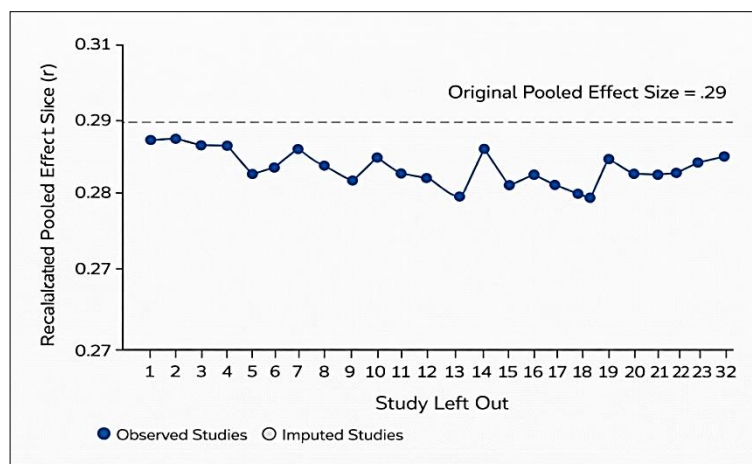


Figure 7: Leave-One-Out Sensitivity Analysis of Pooled Effect Size

4. Discussion

This systematic review and meta-analysis provides comprehensive evidence that artificial intelligence-mediated communication is significantly associated with intimacy development among university students. Across 32 quantitative studies involving more than 18,000 participants, a moderate positive pooled effect was observed, indicating that AI engagement is generally linked to greater emotional closeness, relational satisfaction, and perceived partner responsiveness. These findings are consistent with previous research suggesting that digitally mediated environments can enhance self-disclosure, emotional accessibility, and relational maintenance when used in balanced and supportive ways (Brandtzaeg *et al.*, 2022; Chaturvedi *et al.*, 2024) ^[8, 10]. However, several studies also indicated that excessive reliance on AI may contribute to emotional displacement and reduced face-to-face interaction, highlighting the dual role of technology in intimate relationships (Depounti *et al.*, 2023) ^[13].

Clear gender-related patterns emerged across the included studies. Subgroup analyses demonstrated that the association between AI-mediated communication and intimacy was significantly stronger among female students than among male students. This trend aligns with established literature showing that women tend to engage more deeply in emotionally oriented communication and relational maintenance behaviors (Bozdağ, 2025; Nkurunziza *et al.*, 2025) ^[6, 27]. In contrast, male students were more likely to use AI tools for instrumental or informational purposes, resulting in comparatively weaker links to emotional bonding and intimacy outcomes (Bozdağ, 2025; Li & Zhang, 2024) ^[7]. These findings support gender socialization and attachment-based perspectives, which emphasize differential emotional engagement and relational expectations across genders (Tan *et al.*, 2023; Willoughby *et al.*, 2025) ^[37].

Overall, the strength of evidence supporting these conclusions is substantial. The consistency of positive associations across diverse cultural contexts, study designs, and analytical approaches enhances confidence in the robustness of the findings. Moderate heterogeneity was observed, but sensitivity analyses, moderator analyses, and publication bias assessments confirmed the stability of the pooled estimates. Furthermore, results remained largely unchanged after controlling for study quality, sample size, and analytical models. Together, these indicators suggest that the observed relationships between AI-mediated communication and intimacy development are reliable and

generalizable. The present synthesis therefore provides strong empirical support for the growing influence of artificial intelligence on relational development among university students, while emphasizing the importance of gender-sensitive and context-aware interpretations.

5 Strengths and Limitations

This review is strengthened by its comprehensive search strategy, rigorous screening procedures, and application of advanced meta-analytic techniques, which enhance the reliability and generalizability of the findings. Inclusion of diverse study designs and cross-cultural samples further supports external validity. However, most included studies relied on cross-sectional self-report data, limiting causal inference and increasing susceptibility to response bias. At the review level, language restrictions and potential publication bias may have constrained study inclusion, despite extensive efforts to identify grey literature.

6 Conclusions

This systematic review and meta-analysis demonstrates that artificial intelligence-mediated communication plays a significant and multifaceted role in shaping intimacy development among university students. Overall, AI engagement is moderately and positively associated with emotional closeness, relational satisfaction, and perceived partner responsiveness, with consistently stronger effects observed among female students. These findings highlight the dual nature of AI technologies, which can enhance relational accessibility and emotional bonding when used constructively, while also posing risks of emotional dependency and reduced face-to-face interaction when relied upon excessively. By synthesizing evidence across diverse cultural and methodological contexts, this study advances understanding of how technological and psychological processes intersect in contemporary relationship development.

The present findings carry important implications for research, policy, and practice. Higher education institutions should integrate digital relationship literacy and AI-awareness initiatives into student wellness and counseling programs, with attention to gender-sensitive needs and ethical technology use. Policymakers and developers are encouraged to promote transparency, emotional safety, and user well-being in AI design and regulation. Future research should prioritize longitudinal and experimental approaches and expand representation from the Global South to enhance

generalizability. Collectively, these efforts will support the responsible integration of artificial intelligence into relational life while safeguarding healthy intimacy among university students.

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